CO1 - AT3

Descriptive Written Test

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1. Intelligent Agents in a Smart Home Automation Syetem :-

Introduction :

An intelligent agent is a system that perceives its environment through sensors and acts upon it using actuators to acheive Speific Objectives.

Types of Intelligent Agents :

1) Simple Rifler agent
2) Model-based agent
3) Goal based agent
4) Utility based agent.

Simple Reflex agent : It selects actions based on only the current Percept using Condition - action rules.

model based agent : A model based agent maintains an internal State representing relevant aspects of the environment.

Goal based agent : A goal based agent chooses action that help acheive a predefined goal.

Utility based agent : It evaluvates possible actions using a utility function that measures how desirable each Outcome is.

comparision :

Agent type	Decision making methods	Smart home Applications
Simple reflex	current percept & rules	Motion-based lighting
Model based	Current input & internal State	Occupancy Tracking
Goal based	Actions that achieve goals	Temperature control
Utility-based	Maximizes expected utility	comfort, cost, Security optimization

Most Efficient Agent : Hybrid Approach

A hybrid agent Combining model-based, goal-based, utility based reasoning is most effective for a smart home automation system.

* The model based component maintains information about room occupancy, user habits and environmental Conditions.

* The goal based component work towards objectives such as maintaining Comfortable temperatures & securing the house.

* The utility based component balances Comfort, energy Consumption, and safety to select the best action.

Example :

When the house becomes empty, the system identifies the absence of occupants, switches off unnecessary lights, adjusts the air conditioner to energy saving setting, and activates security monitoring.

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2. Uninformed Search Strategies in an Unknown Maze

* Uninformed Search, also called blind search, explores possible states without using additional knowledge about how close the state is to a goal.

* The robot can use Uniform cost search (UCS) or Iterative deepening search (IDS) to find a path to exit.

Uniform Cost Search :-

Uniform Cost search expands the node with lowest cumulative path cost, $g(n)$. It uses a priority queue to select the least-cost unexplored node.

Working procedure :-

1. Place the starting position at a priority queue with cost zero.

2. Remove the node with the lowest cumulative cost.

3. Check whether the node represent the exit. If so, return the path.

4. Otherwise, explore its valid neighbouring cells & calculate their cumulative costs.

5. Add new or cheaper paths to the priority queue.

6. Repeat until the exit is found or no reachable nodes remain.

2. Iterative Deepening Search (IDS):

Iterative Deepening Search repeatedly performs depth-limited depth first-search, increasing the depth limit until the goal is found.

Working Procedure:

1. Set the depth limit to zero.
2. Search the maze from the starting position up to that limit.
3. If the exit is not found, increase the limit by one.
4. Repeat the depth-limited search
5. Stop when the exit is reached.

Example:

The robot first checks the starting cell, the reaches up to depth 1, depth 2, depth 3 and do on until the exit is reached.

Comparision of UCS & IDS

Feature	Uniform cost search	IDS
Search method	Lowest Cumulative cost first	Increasing depth limits
Data structure	Priority queue	Depth-first Search stack
Complete	Yes, under positive lower bounded step costs	Yes, with finite branching.
Optimality	Minimum cost-path	Shallowest path for equal step costs
Time Complexity	Exponential in the general unit-cost case	$O(b^d)$
Space Complexity	Exponential in the general Unit cost case	$O(bd)$
Best suited for	Different movement costs	limited memory & equal movement costs.

Most effective Combination :-

For an unknown maze with limited memory and equal movement costs, a model-based, goal-based agent using IDS is an effective choice.

* The model-based Component remembers discovered walls & visited locations.

* The goal-based Component directs the search toward the exit.

* IDS gradually increases the search depth while using limited memory.

* Recording visited states and the current search path helps prevent repeated exploration and unnecessary movement.